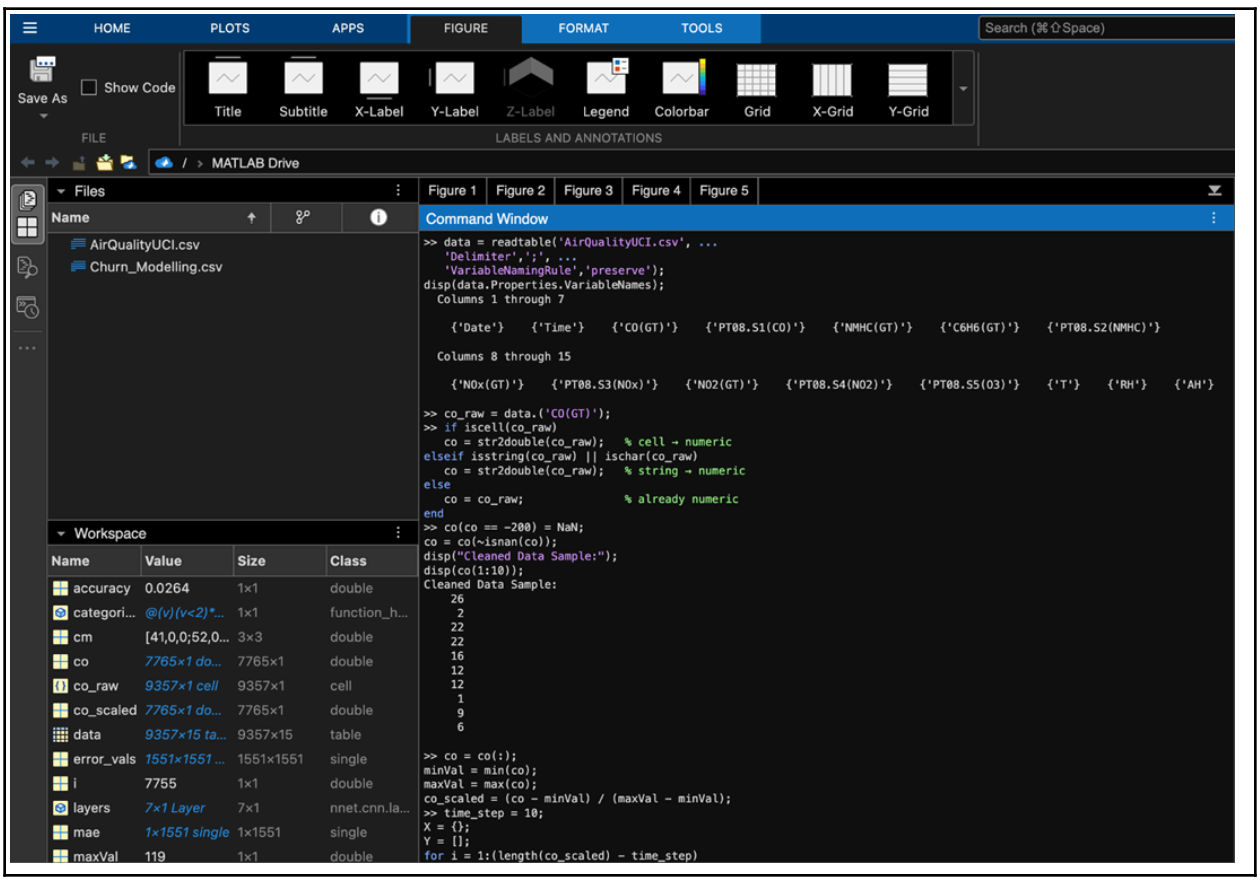
Experiment 9

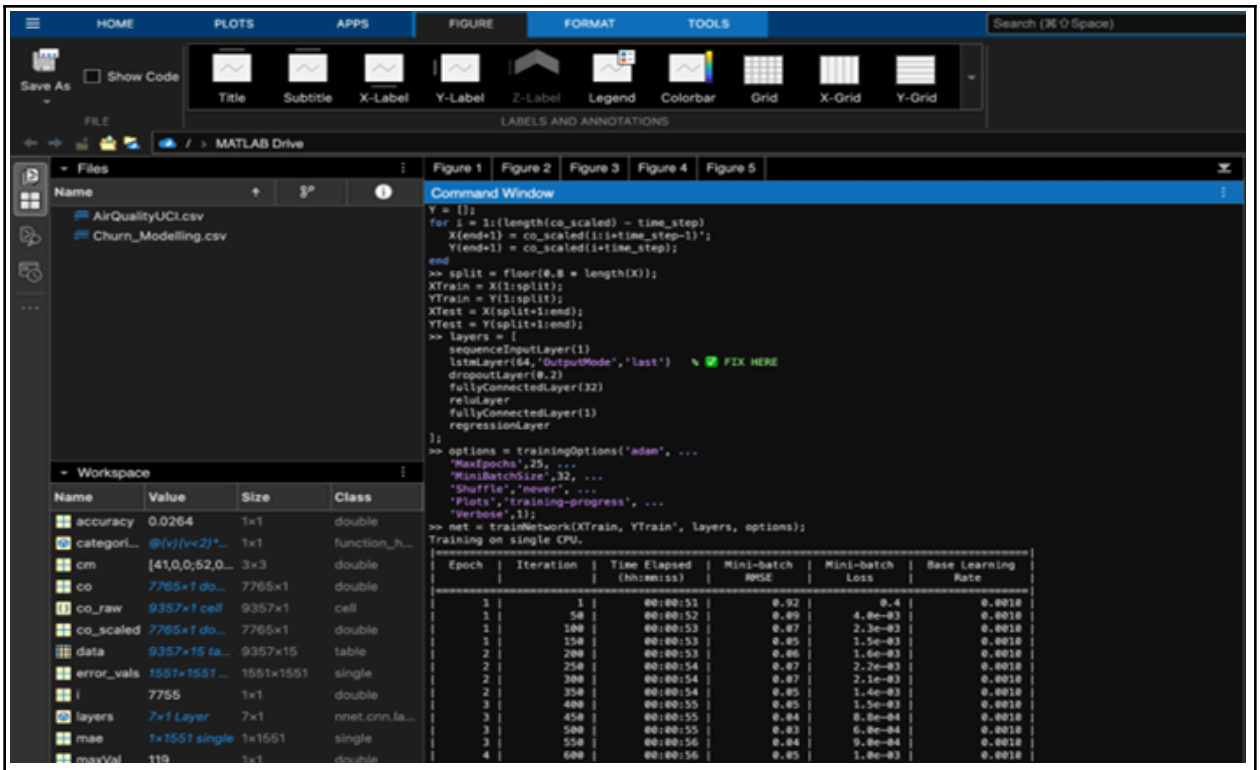
Aim : Implement Recurrent Neural Network (RNN) / LSTM for time series or text data

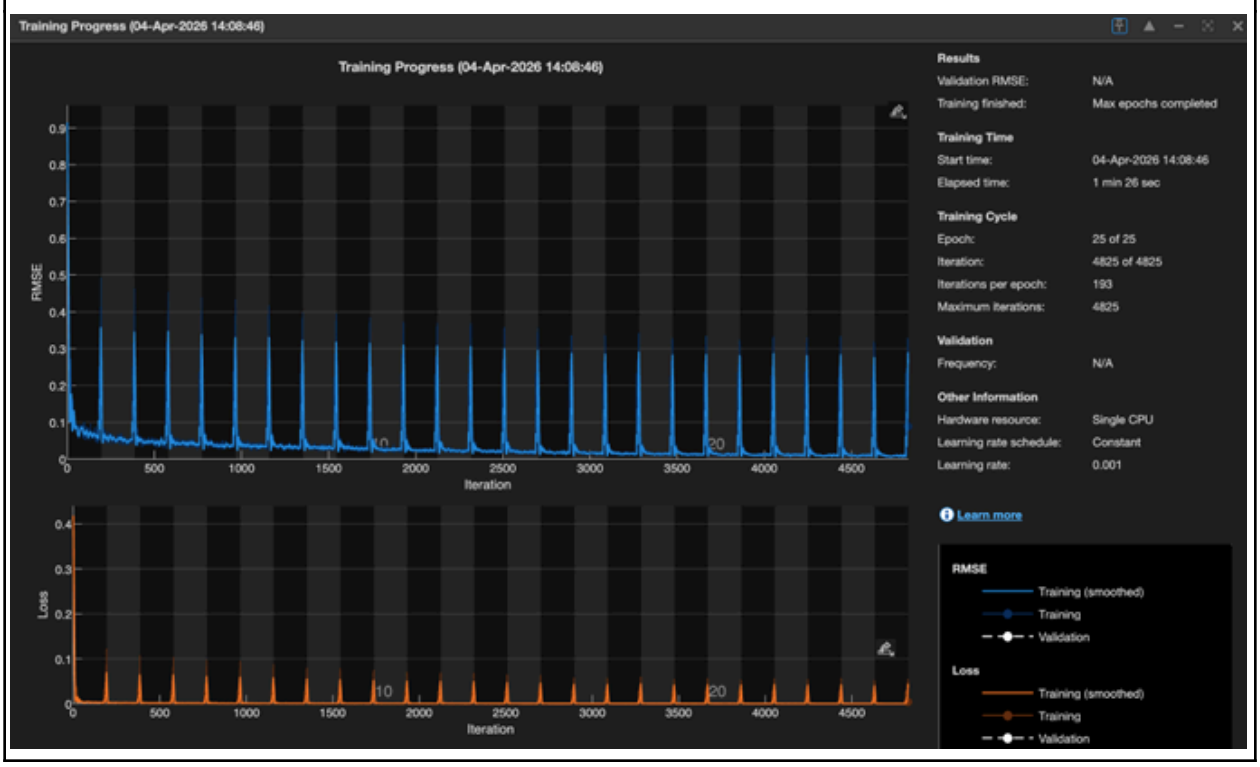
Data Source :

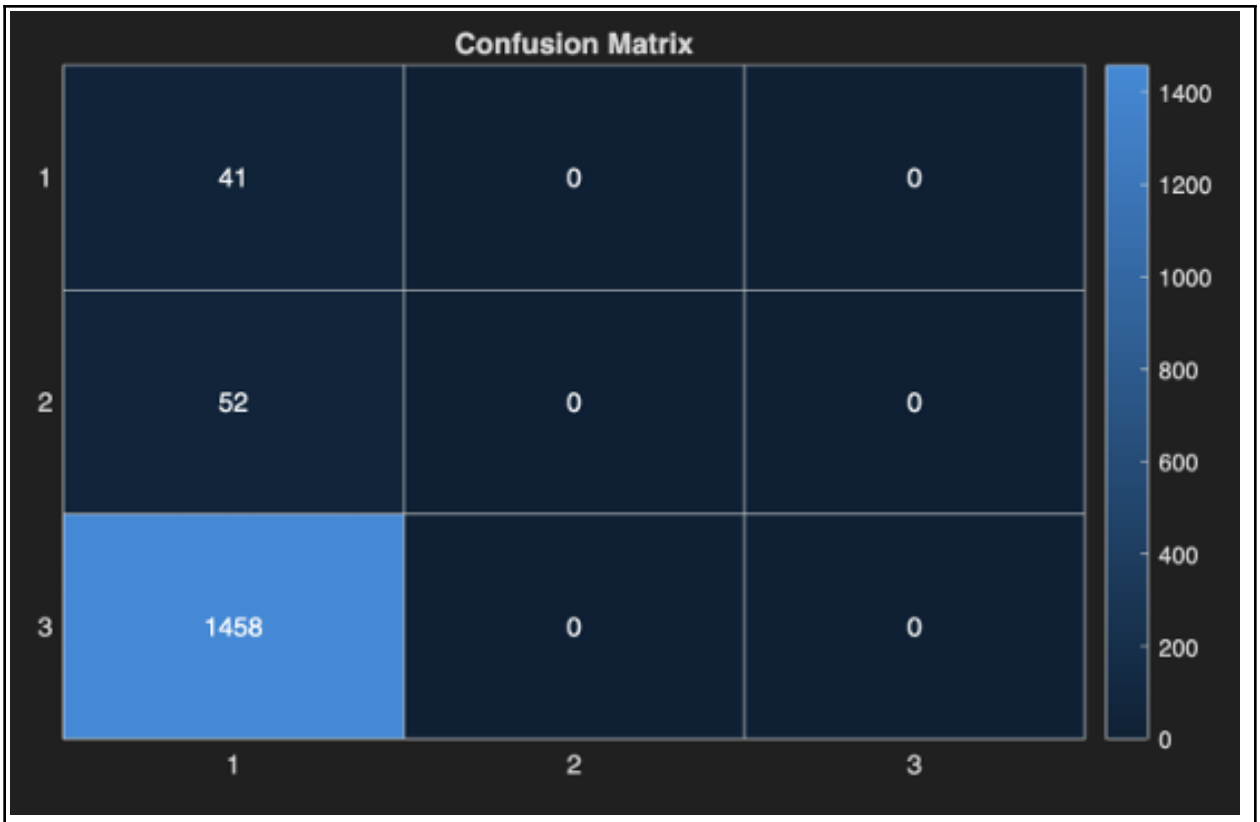
- 9,358 instances with 15 features (time-series air quality data)
- Collected hourly from March 2004 to February 2005
- Based on 5 chemical sensors with ground truth values
- Measures pollutants like CO, NO_2, Benzene, and NOx
- Contains missing values (-200) and sensor noise issues
- [DataSet Link]

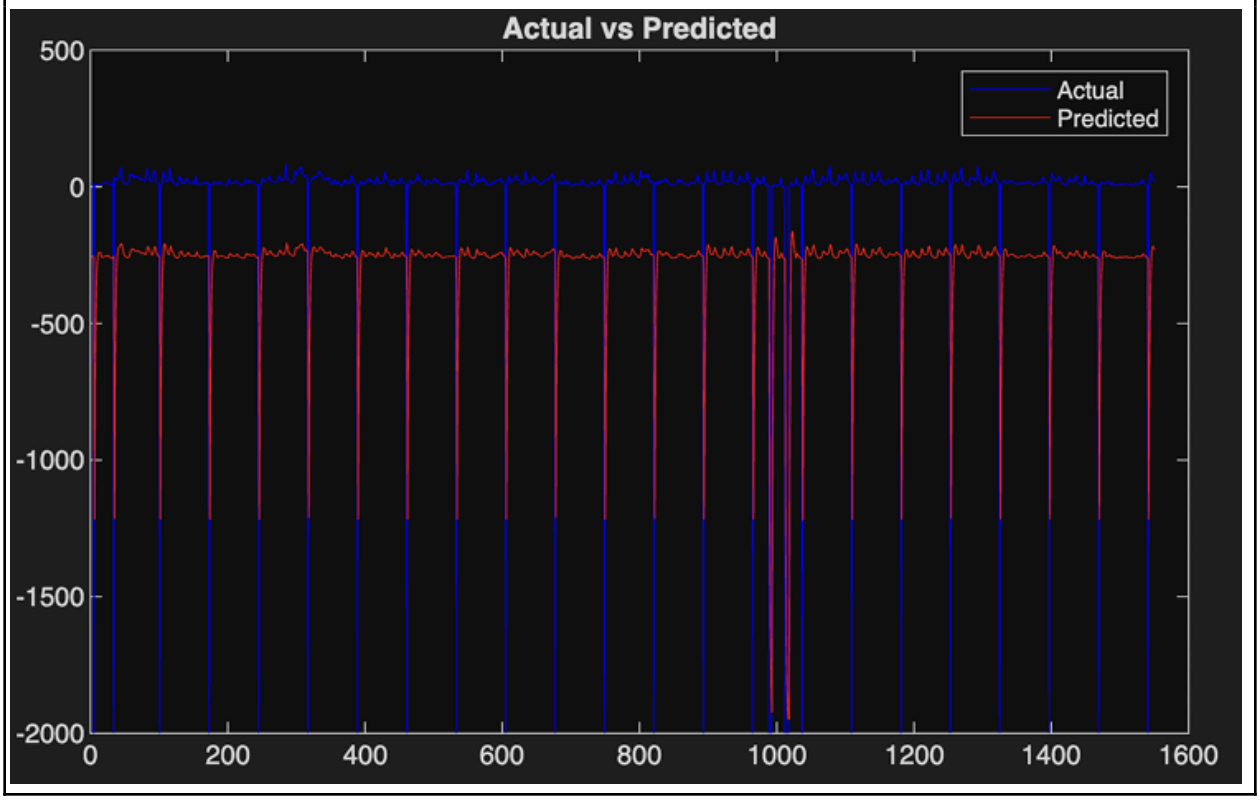
Output :

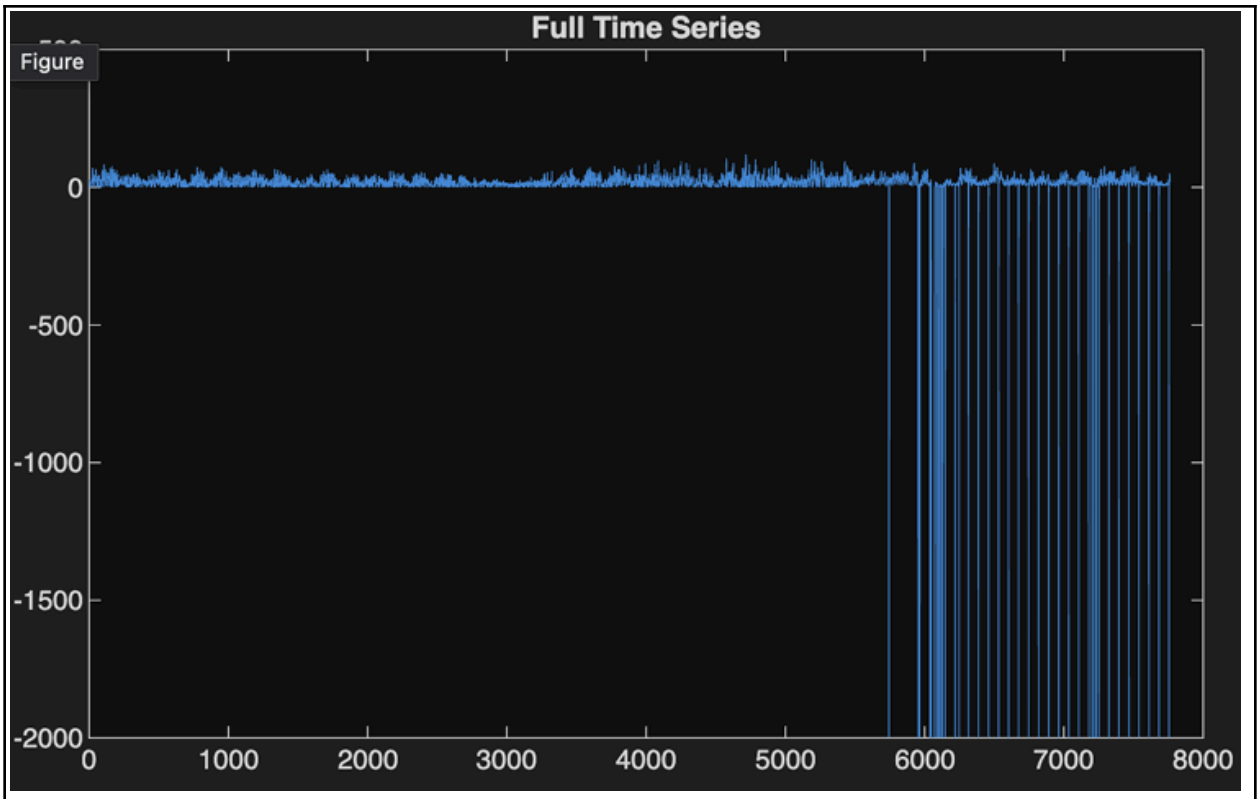


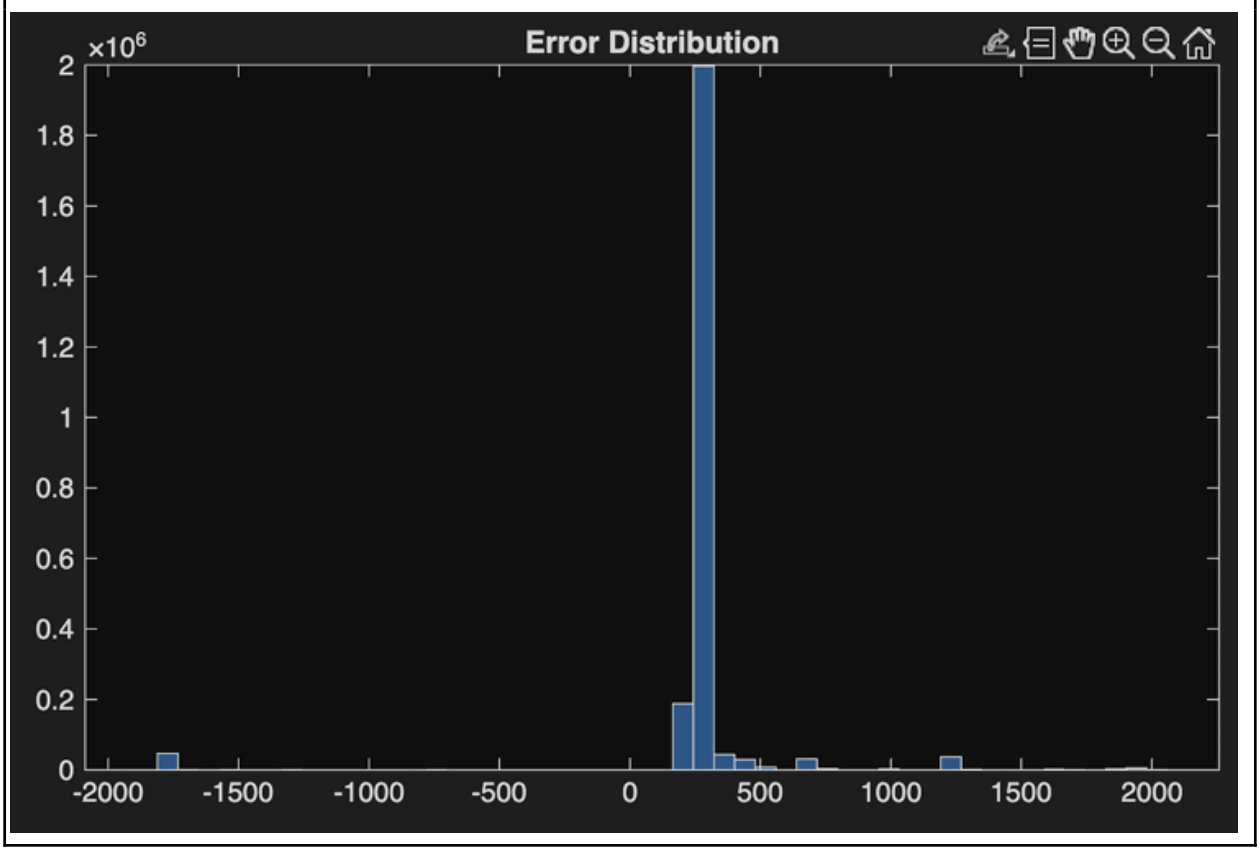

HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Save As
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
AirQualityUCI.csv
Churn_Modelling.csv
Figure 1
Figure 2
Figure 3
Figure 4
Figure 5
Command Window
Workspace
Name
Value
Size
Class
accuracy
0.0264
1×1
double
cm
[41,0,0;52,0...
3×3
double
co
7765×1 do...
7765×1
double
co_raw
9357×1 cell
9357×1
cell
co_scaled
7765×1 do...
7765×1
double
data
9357×15 ta...
9357×15
table
error_vals
1551×1551 ...
1551×1551
single
i
7755
1×1
double
layers
7×1 Layer
7×1
nnet.cnn.la...
mae
1×1551 single
1×1551
single
Y = [];
for i = 1:(length(co_scaled) - time_step)
X(end+1) = co_scaled(i:i+time_step-1)';
Y(end+1) = co_scaled(i+time_step);
end
>> split = floor(0.8 * length(X));
XTrain = X(1:split);
YTrain = Y(1:split);
XTest = X(split+1:end);
YTest = Y(split+1:end);
>> layers = [
sequenceInputLayer(1)
lstmLayer(64,'OutputMode','last')
FIX HERE
dropoutLayer(0.2)
fullyConnectedLayer(32)
reluLayer
fullyConnectedLayer(1)
regressionLayer
];
>> options = trainingOptions('adam', ...
'MaxEpochs',25, ...
'MiniBatchSize',32, ...
'Shuffle','never', ...
'Plots','training-progress', ...
'Verbose',1);
>> net = trainNetwork(XTrain, YTrain', layers, options);
Training on single CPU.
Epoch	Iteration	Time Elapsed (hh:mm:ss)	Mini-batch RMSE	Mini-batch Loss	Base Learning Rate
1	1	00:00:51	0.92	0.4	0.0010
1	50	00:00:52	0.09	4.0e-03	0.0010
1	100	00:00:53	0.07	2.3e-03	0.0010
1	150	00:00:53	0.05	1.5e-03	0.0010
2	200	00:00:53	0.06	1.6e-03	0.0010
2	250	00:00:54	0.07	2.2e-03	0.0010
2	300	00:00:54	0.07	2.1e-03	0.0010
2	350	00:00:54	0.05	1.4e-03	0.0010
3	400	00:00:55	0.05	1.5e-03	0.0010
3	450	00:00:55	0.04	8.0e-04	0.0010
3	500	00:00:55	0.03	6.0e-04	0.0010
3	550	00:00:56	0.04	9.0e-04	0.0010
4	600	00:00:56	0.05	1.0e-03	0.0010


Training Progress (04-Apr-2026 14:08:46)
Training Progress (04-Apr-2026 14:08:46)
RMSE
Iteration
Loss
Results
Validation RMSE: N/A
Training finished: Max epochs completed
Training Time
Start time: 04-Apr-2026 14:08:46
Elapsed time: 1 min 26 sec
Training Cycle
Epoch: 25 of 25
Iteration: 4825 of 4825
Iterations per epoch: 193
Maximum iterations: 4825
Validation
Frequency: N/A
Other Information
Hardware resource: Single CPU
Learning rate schedule: Constant
Learning rate: 0.001
Learn more
RMSE
Training (smoothed)
Training
Validation
Loss
Training (smoothed)
Training
Validation


Confusion Matrix
1
2
3
41
0
0
52
0
0
1458
0
0
1
2
3
1400
1200
1000
800
600
400
200
0


Actual vs Predicted
Actual
Predicted
500
0
-500
-1000
-1500
-2000
0
200
400
600
800
1000
1200
1400
1600


Full Time Series
Figure
0
-500
-1000
-1500
-2000
0
1000
2000
3000
4000
5000
6000
7000
8000


Error Distribution
×10^6
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
-2000
-1500
-1000
-500
0
500
1000
1500
2000

Conclusion

RNN and LSTM are important models for sequence learning tasks. While RNN is suitable for short-term dependencies, LSTM is preferred for problems involving long-term dependencies, making it widely used in time series analysis and natural language processing.